Algorithms

Lecture #20

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Chain Matrix Multiplication

Chain Matrix Multiplication

Dynamic Programming Formulation

- We will store the solutions to the subproblem in a table and build the table bottom-up (why)?
- For $1 \leq i \leq j \leq n$, let m[i, j] denote the minimum number of multiplications needed to compute Ai..j.

Chain Matrix Multiplication

Chain Matrix Multiplication

Chain Matrix Multiplication

- For $1 \leq i \leq j \leq n$, let $m[i, j]$ denote the minimum number of multiplications needed to compute $A_{i..j}$.
- The optimum can be described by the following recursive formulation.

Chain Matrix Multiplication

Chain Matrix Multiplication

- The optimum time to compute $A_{i..k}$ is $m[i, k]$ and optimum time for $A_{k+1..j}$ is in $m[k + 1, j]$.
- Since $A_{i..k}$ is a $p_{i-1} \times p_k$ matrix and $A_{k+1..j}$ is $p_k \times p_j$ matrix, the time to multiply them is $p_{i-1} \times p_k \times p_j$.

Chain Matrix Multiplication

Chain Matrix Multiplication

- This suggests the following recursive rule:

$$m[i, i] = 0$$

$$m[i,j] = \min_{i \le k < j} \left(m[i,k] + m[k+1,j] + p_{i-1} p_k p_j \right)$$

Chain Matrix Multiplication

Chain Matrix Multiplication

- We do not want to calculate m entries recursively.
- How should we proceed?
- We will fill m along the diagonals. Here is how.

Chain Matrix Multiplication

Chain Matrix Multiplication

- Set all m[i, i] = 0 using the base condition.
- Compute cost for multiplication of a sequence of 2 matrices.
- These are m[1,2],m[2,3],m[3,4],...,m[n-1,n].
- m[1, 2], for example is

  $m[1,2] = m[1,1] + m[2,2] + p_0 \cdot p_1 \cdot p_2$

Chain Matrix Multiplication

Chain Matrix Multiplication

For example, for m for product of 5 matrices at this stage would be:

m [1,1]	←m [1,2] ↓			
	m [2,2]	←m [2,3] ↓		
		m [3,3]	←m [3,4] ↓	
			m [4,4]	←m [4,5] ↓
				m [5,5]

Chain Matrix Multiplication

Chain Matrix Multiplication

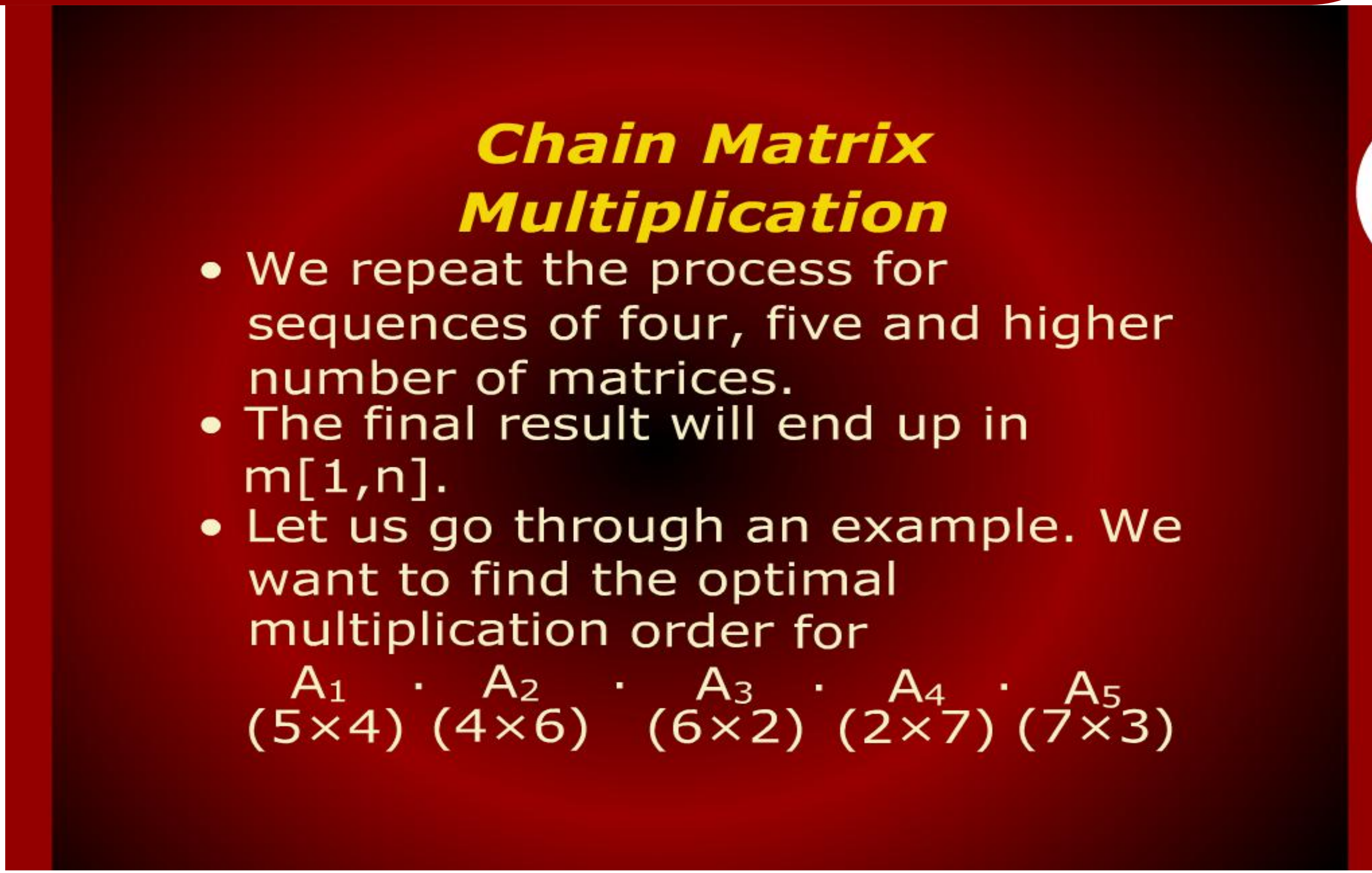
- Next, we compute cost of multiplication for sequences of three matrices.
- These are

$m[1,3], m[2,4], m[3,5], \ldots, m[n-2,n]$.

- $m[1, 3]$, for example is

$$m[1,3] = \min \begin{cases} m[1,1] + m[2,3] + p_0 \cdot p_1 \cdot p_3 \\ m[1,2] + m[3,3] + p_0 \cdot p_2 \cdot p_3 \end{cases}$$

Chain Matrix Multiplication



Chain Matrix Multiplication

Chain Matrix Multiplication

Initially

0				
	0			
		0		
			0	
				0

Chain Matrix Multiplication

Chain Matrix Multiplication

$$m[1,2] = m[1,1] + m[2,2] + p_0 \cdot p_1 \cdot p_2$$
$$= 0 + 0 + 5 \cdot 4 \cdot 6 = 120$$

$$m[2,3] = m[2,2] + m[3,3] + p_1 \cdot p_2 \cdot p_3$$
$$= 0 + 0 + 4 \cdot 6 \cdot 2 = 48$$

Chain Matrix Multiplication

Chain Matrix Multiplication

$$m[3,4] = m[3,3] + m[4,4] + p_2 \cdot p_3 \cdot p_4$$
$$= 0 + 0 + 6 \cdot 2 \cdot 7 = 84$$

$$m[4,5] = m[4,4] + m[5,5] + p_3 \cdot p_4 \cdot p_5$$
$$= 0 + 0 + 2 \cdot 7 \cdot 3 = 42$$

Chain Matrix Multiplication

Chain Matrix Multiplication

after product of 2 matrices (first super diagonal)

0	120			
	0	48		
		0	84	
			0	42
				0

Chain Matrix Multiplication

Chain Matrix Multiplication

$$m[1,3] = m[1,1] + m[2,3] + p_0 \cdot p_1 \cdot p_3$$
$$= 0 + 48 + 5 \cdot 4 \cdot 2 = 88$$

$$m[1,3] = m[1,2] + m[3,3] + p_0 \cdot p_2 \cdot p_3$$
$$= 120 + 0 + 5 \cdot 6 \cdot 2 = 180$$

minimum $m[1,3] = 88$ at $k = 1$

Chain Matrix Multiplication

Chain Matrix Multiplication

$m[2,4] = m[2,2] + m[3,4] + p_1 \cdot p_2 \cdot p_4$

$= 0 + 84 + 4 \cdot 6 \cdot 7 = 252$

$m[2,4] = m[2,3] + m[4,4] + p_1 \cdot p_3 \cdot p_4$

$= 48 + 0 + 4 \cdot 2 \cdot 7 = 104$

minimum $m[2,4] = 104$ at $k = 3$

Chain Matrix Multiplication

Chain Matrix Multiplication

$$m[3,5] = m[3,3] + m[4,5] + p_2 \cdot p_3 \cdot p_5$$
$$= 0 + 42 + 6 \cdot 2 \cdot 3 = 78$$

$$m[3,5] = m[3,4] + m[5,5] + p_2 \cdot p_4 \cdot p_5$$
$$= 84 + 0 + 6 \cdot 7 \cdot 3 = 210$$

minimum $m[3, 5] = 78$ at $k = 3$

Chain Matrix Multiplication

Chain Matrix Multiplication

product of 3 matrices, (second super diagonal)

0	120	88		
	0	48	104	
		0	84	78
			0	42
				0

Chain Matrix Multiplication

Chain Matrix Multiplication

$$m[1,4] = m[1,1] + m[2,4] + p_0 \cdot p_1 \cdot p_4$$
$$= 0 + 104 + 5 \cdot 4 \cdot 7 = 244$$

$$m[1,4] = m[1,2] + m[3,4] + p_0 \cdot p_2 \cdot p_4$$
$$= 120 + 84 + 5 \cdot 6 \cdot 7 = 414$$

$$m[1,4] = m[1,3] + m[4,4] + p_0 \cdot p_3 \cdot p_4$$
$$= 88 + 0 + 5 \cdot 2 \cdot 7 = 158$$

minimum $m[1, 4] = 158$ at $k = 3$

Chain Matrix Multiplication

Chain Matrix Multiplication

$$m[2,5] = m[2,2] + m[3,5] + p_1 \cdot p_2 \cdot p_5$$
$$= 0 + 78 + 4 \cdot 6 \cdot 3 = 150$$

$$m[2,5] = m[2,3] + m[4,5] + p_1 \cdot p_3 \cdot p_5$$
$$= 48 + 42 + 4 \cdot 2 \cdot 3 = 114$$

$$m[2,5] = m[2,4] + m[5,5] + p_1 \cdot p_4 \cdot p_5$$
$$= 104 + 0 + 4 \cdot 7 \cdot 3 = 188$$

minimum $m[2,5] = 114$ at $k = 3$

Chain Matrix Multiplication

Chain Matrix Multiplication

Product of 4 matrices

0	120	88	158	
	0	48	104	114
		0	84	78
			0	42
				0

Chain Matrix Multiplication

Chain Matrix Multiplication

$$m[1,5] = m[1,1] + m[2,5] + p_0 \cdot p_1 \cdot p_5$$
$$= 0 + 114 + 5 \cdot 4 \cdot 3 = 174$$
$$m[1,5] = m[1,2] + m[3,5] + p_0 \cdot p_2 \cdot p_5$$
$$= 120 + 78 + 5 \cdot 6 \cdot 3 = 288$$
$$m[1,5] = m[1,3] + m[4,5] + p_0 \cdot p_3 \cdot p_5$$
$$= 88 + 42 + 5 \cdot 2 \cdot 3 = 160$$
$$m[1,5] = m[1,4] + m[5,5] + p_0 \cdot p_4 \cdot p_5$$
$$= 158 + 0 + 5 \cdot 7 \cdot 3 = 263$$

minimum $m[1,5] = 160$ at $k=3$

Chain Matrix Multiplication

Chain Matrix Multiplication

Final cost matrix

0	120	88	158	160
0	0	48	104	114
0	0	0	84	78
0	0	0	0	42
0	0	0	0	0

Chain Matrix Multiplication

Chain Matrix Multiplication

Order in which m entries are calculated

0	1	5	8	10
0	0	2	6	9
0	0	0	3	7
0	0	0	0	4
0	0	0	0	0

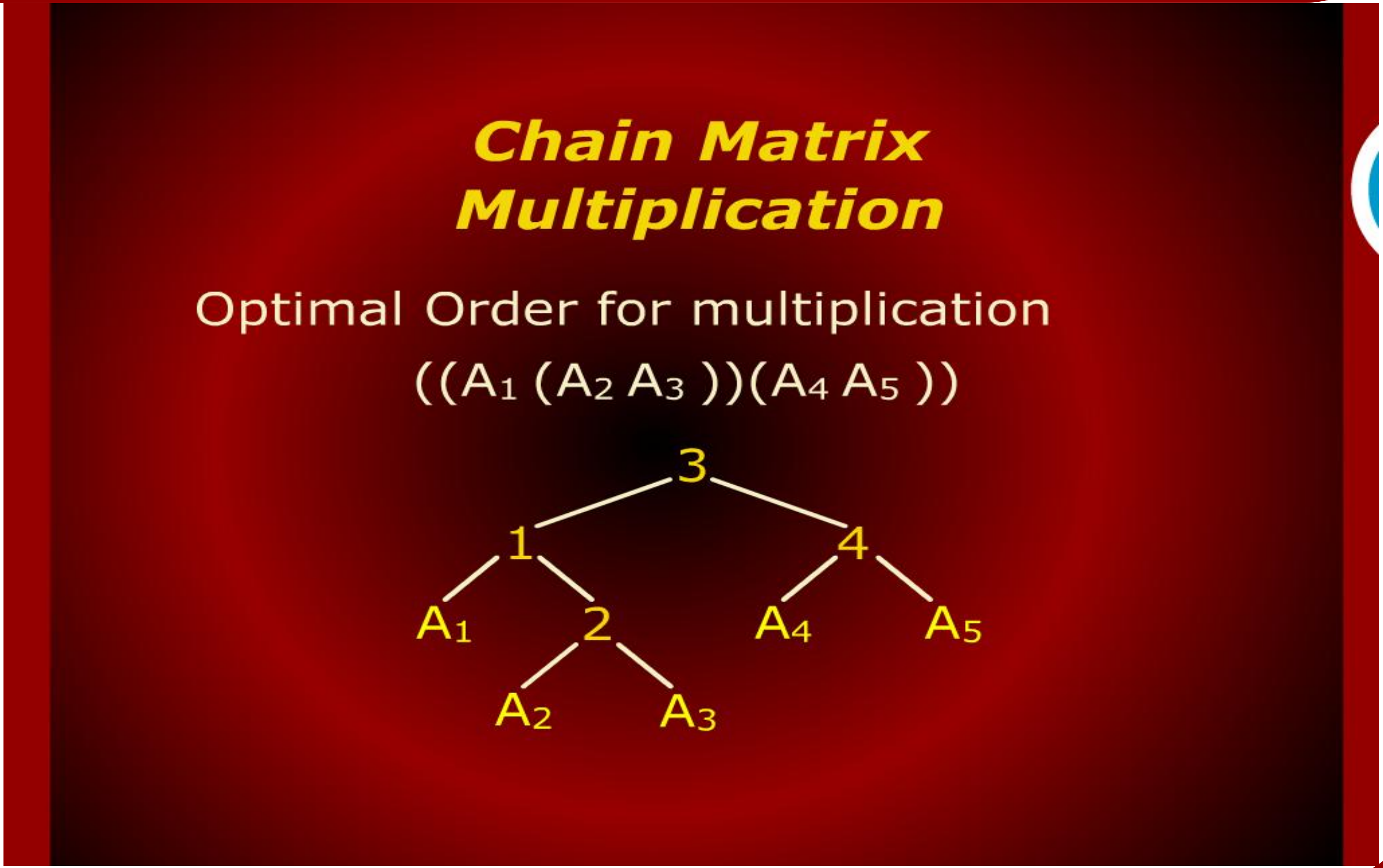
Chain Matrix Multiplication

Chain Matrix Multiplication

Split k values

0	1	1	3	3
	0	2	3	3
		0	3	3
			0	4
				0

Chain Matrix Multiplication



Chain Matrix Multiplication

Chain Matrix Multiplication

- If $i = j$, there is only one matrix and thus $m[i, i] = 0$ (the diagonal entries).
- If $i < j$, the we are asking for the product $A_{i..j}$.
- This can be split by considering each k, $i \le k < j$, as $A_{i..k}$ times $A_{k+1..j}$.